

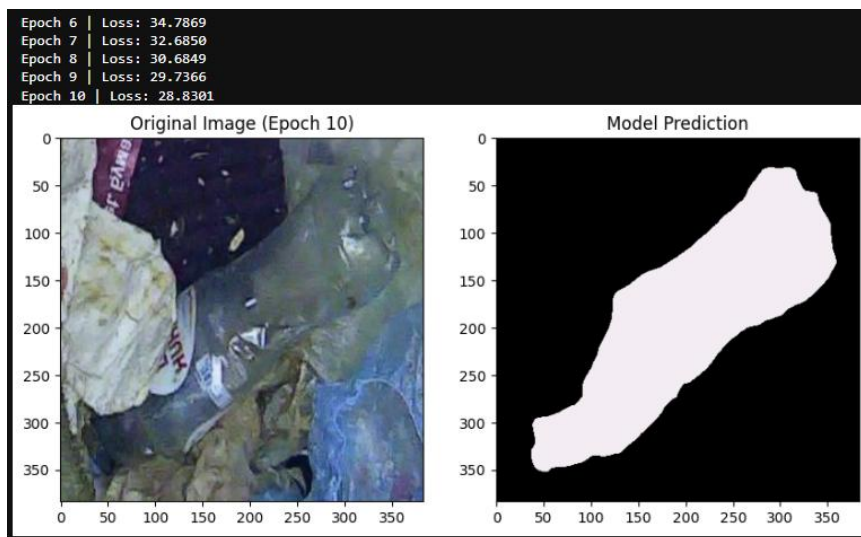
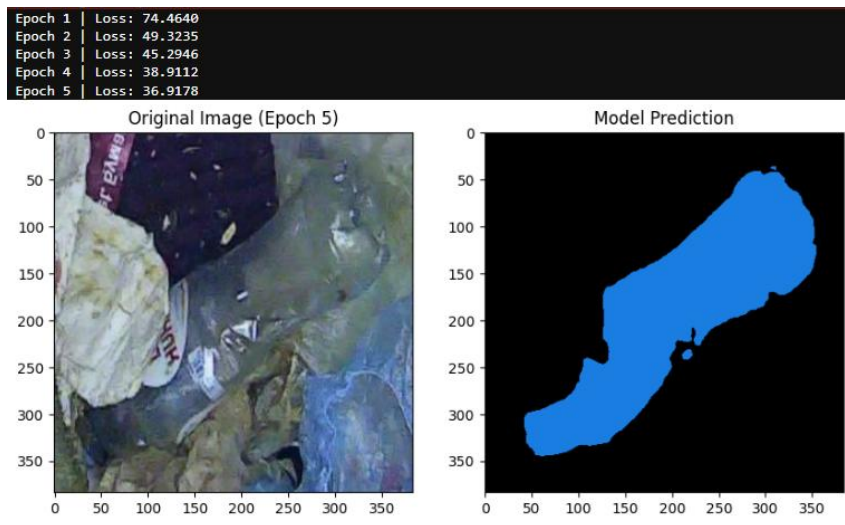
Mask2Former Experiments & Results

Experiment 1 – initial trial

Hyperparameters:

- Learning Rate (lr): $5e-5$
- Weight Decay: 0.05
- Batch Size: 2
- Epochs: 10

Loss evolution + visual checks at every 5 epochs:



Metric Results:

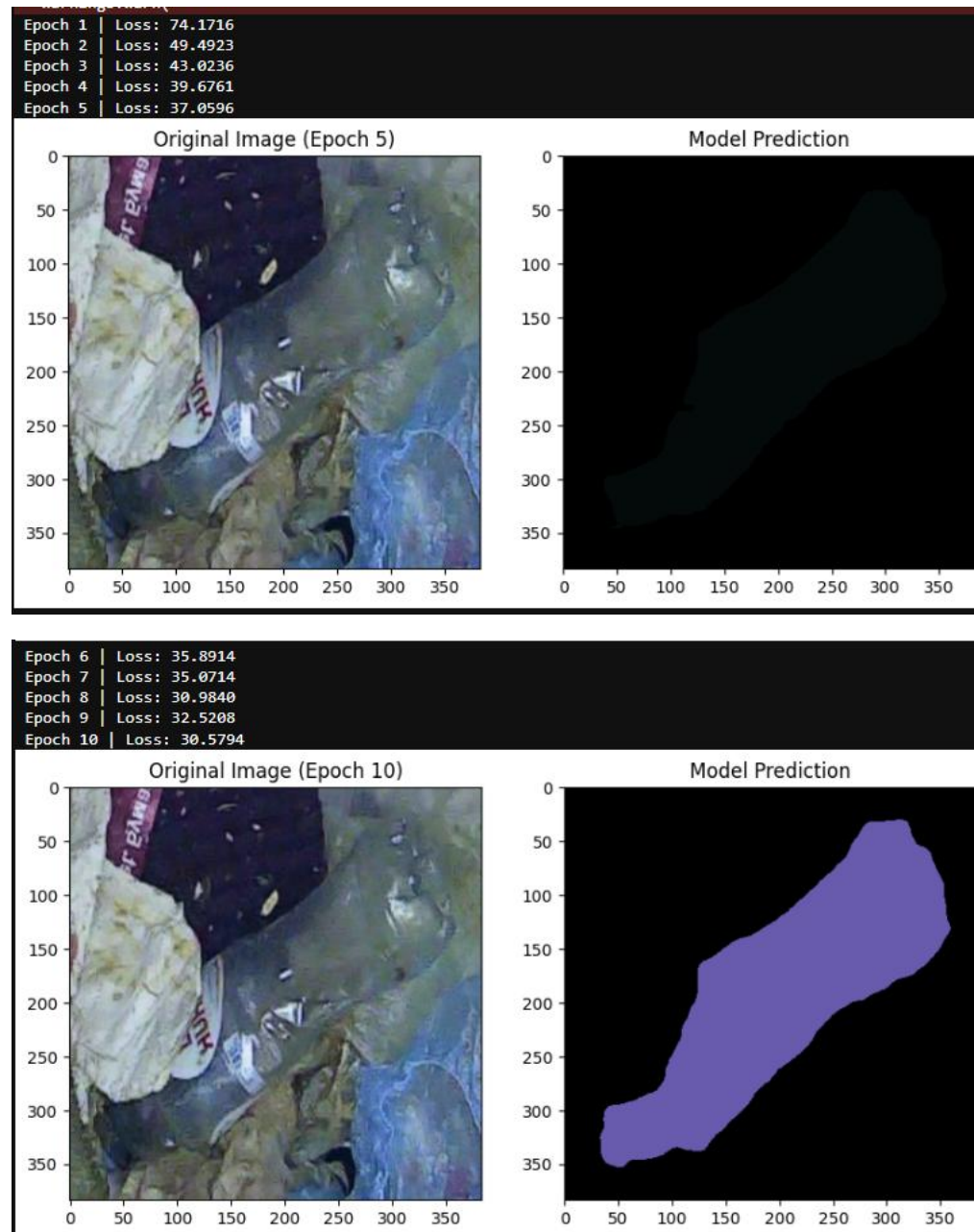
Mean IoU (mIoU):	0.3978
Mean Precision:	0.6124
Mean Recall:	0.6464
Mean F1-Score:	0.6144

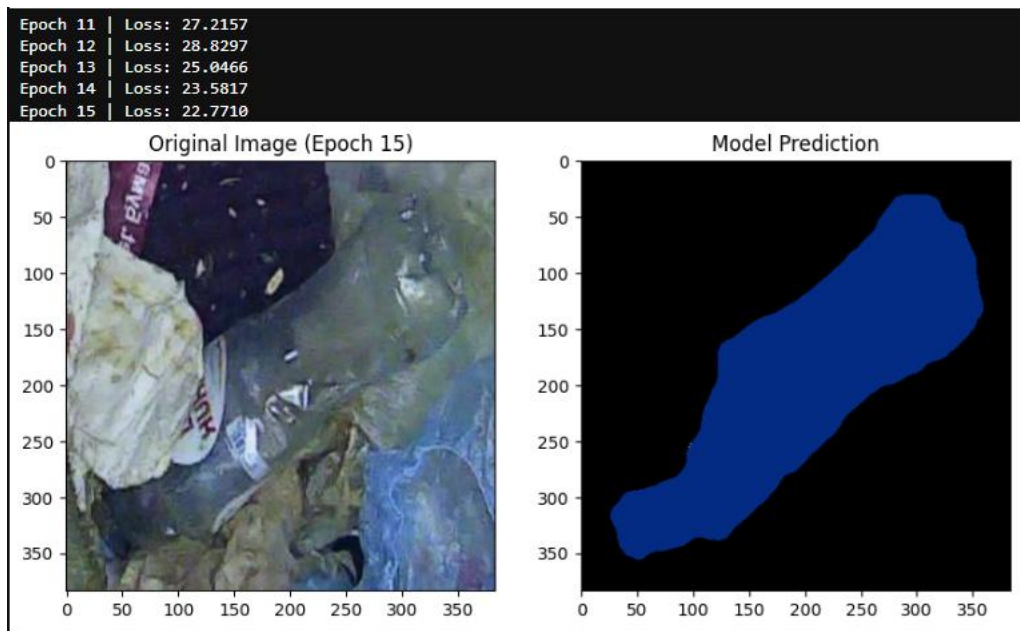
Experiment 2 – epochs increased

Hyperparameters:

- Learning Rate (lr): $5e-5$
- Weight Decay: 0.05
- Batch Size: 2
- Epochs: 15

Loss evolution + visual checks at every 5 epochs:





Metrics Results:

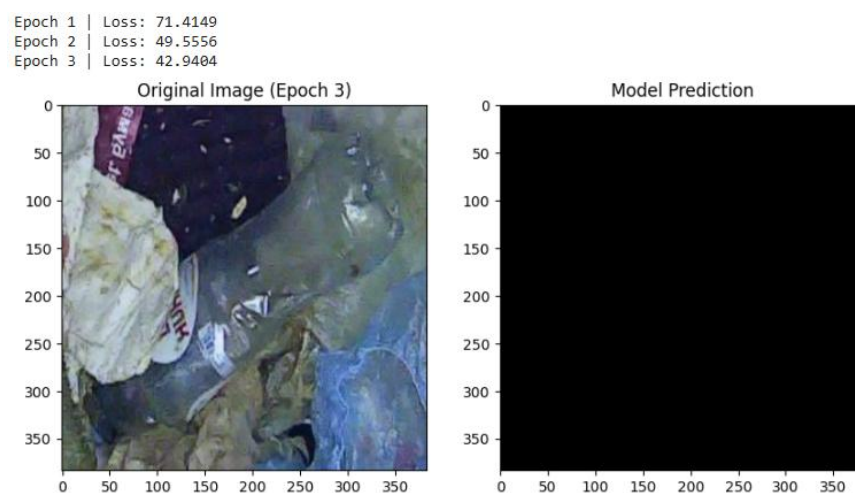
Mean IoU (mIoU):	0.4605
Mean Precision:	0.6970
Mean Recall:	0.6883
Mean F1-Score:	0.6834

Experiment 3: Epochs + weight decay increased + predictions visualized every 3 epochs

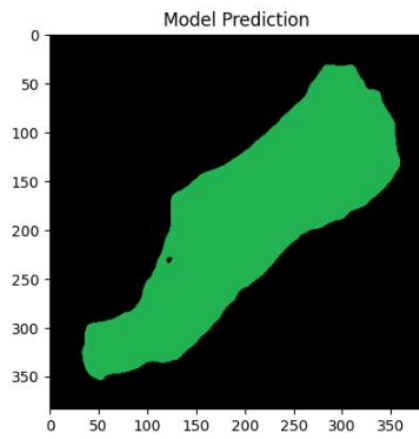
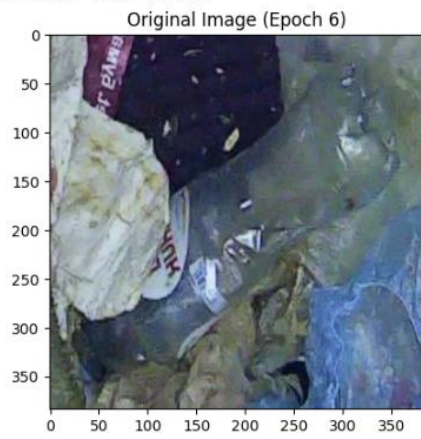
Hyperparameters:

- Learning Rate (lr): $5e-5$
- Weight Decay: 0.1
- Batch Size: 2
- Epochs: 18

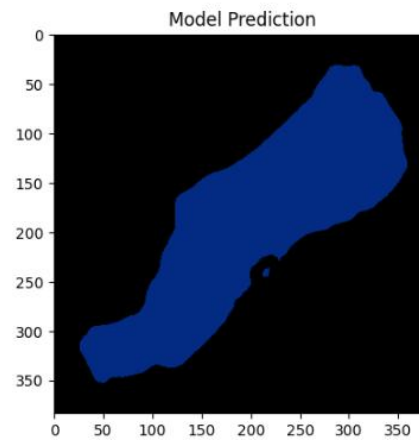
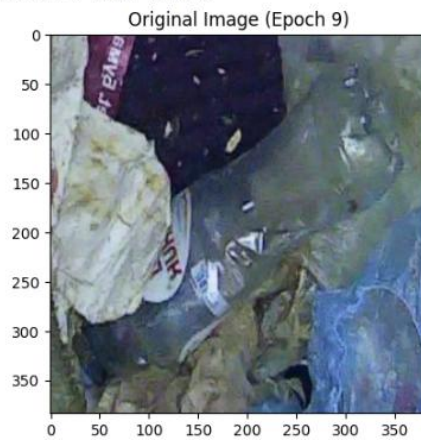
Loss evolution + visual checks at every 3 epochs:



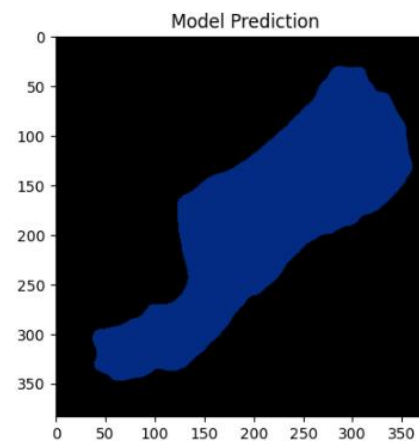
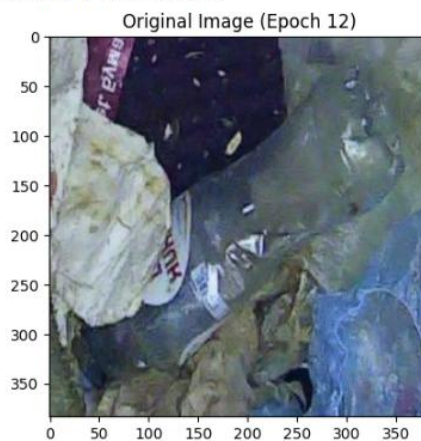
Epoch 4 | Loss: 37.8729
Epoch 5 | Loss: 35.2764
Epoch 6 | Loss: 33.4926



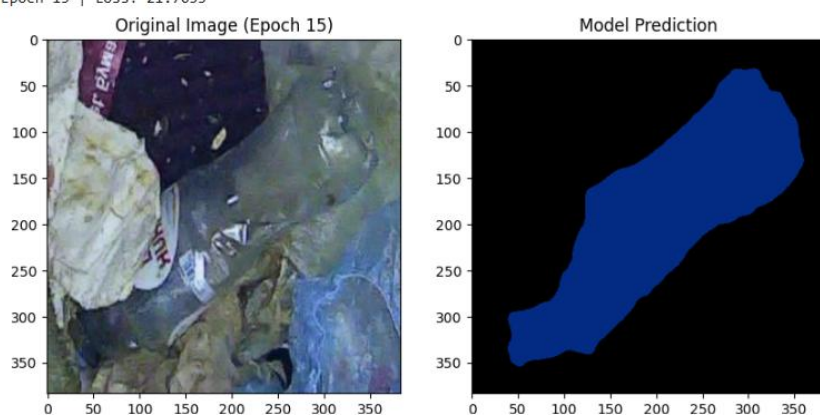
Epoch 7 | Loss: 32.7424
Epoch 8 | Loss: 30.0226
Epoch 9 | Loss: 29.7055



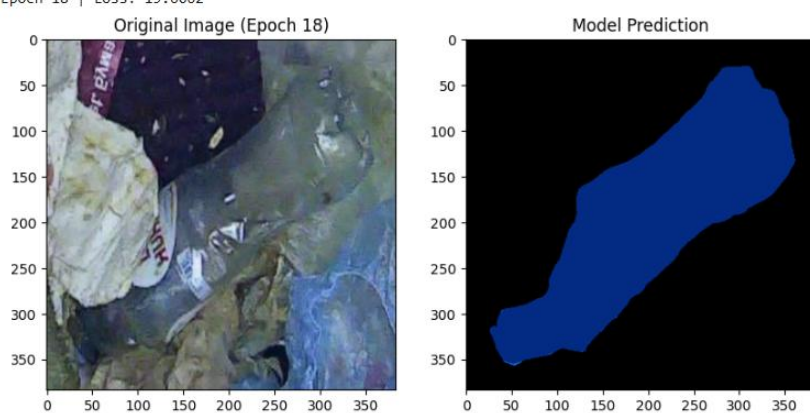
Epoch 10 | Loss: 28.0875
Epoch 11 | Loss: 25.7786
Epoch 12 | Loss: 25.1878



Epoch 13 | Loss: 23.5042
Epoch 14 | Loss: 22.4787
Epoch 15 | Loss: 21.7053



Epoch 16 | Loss: 21.7626
Epoch 17 | Loss: 21.1163
Epoch 18 | Loss: 19.6062



Metrics Results:

Mean IoU (mIoU): 0.7194
Mean Precision: 0.8731
Mean Recall: 0.8702
Mean F1-Score: 0.8651

The model was also used to evaluate and visualise some random samples. Below are some of the samples:



Original Waste Image



Mask2Former Prediction Overlay



Original Waste Image



Mask2Former Prediction Overlay

